

Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Whole Blood EDTA

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:55AM
Report Generated On : 12/Sep/2026 02:11PM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF BIOCHEMISTRY HBA1C

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

HbA1c - Glycosylated Hemoglobin

HbA1c (Glycosylated Hemoglobin)	5.20	%	4.2 - 5.7
Method: HPLC			
Machine: BIORAD D100			
Average Estimated Glucose	102.54	mg/dl	
Method: Calculated			

INTERPRETATION:

AS PER AMERICAN DIABETES ASSOCIATION (ADA):

REFERENCE GROUP

Non diabetic
AT Risk (Prediabetes)
Diagnosing Diabetes

GLYCOSYLATED HEMOGLOBIN (HBA1c) IN %

<5.7

5.7 – 6.4

>= 6.5

Age > 19 Years

Goals of Therapy: < 7.0

Actions Suggested: >8.0

Age < 19 Years

Goal of therapy: <7.5

Therapeutic goals FOR glycemic control

REMARKS

- HbA1c IS used FOR monitoring diabetic control. It reflects the mean plasma glucose over three months
 - HbA1c may be falsely low IN diabetics WITH hemolytic disease. IN these individuals a plasma fructosamine LEVEL may be used which evaluates diabetes over 15 days.
 - Inappropriately low HbA1c VALUES may be reported due TO hemolysis, recent blood transfusion, acute blood loss, hypertriglyceridemia, chronic liver disease. Drugs LIKE dapsone, ribavirin, antiretroviral drugs, trimethoprim, may also cause interference WITH estimation of HbA1c, causing falsely low values.
 - HbA1c may be increased IN patients WITH polycythemia OR post-splenectomy.
 - Inappropriately higher VALUES of HbA1c may be caused due TO iron deficiency, vitamin B12 deficiency, alcohol intake, uremia, hyperbilirubinemia AND large doses of aspirin.
 - Trends IN HbA1c are a better indicator of diabetic control THAN a solitary test.
 - ANY sample WITH >15% HbA1c should be suspected OF HAVING a hemoglobin variant, especially IN a non-diabetic patient. Similarly, below 4% should prompt additional studies TO determine the possible presence of variant hemoglobin.
 - HbA1c target IN pregnancy IS TO attain LEVEL <6 % .
 - HbA1c target IN paediatric age GROUP IS TO attain LEVEL < 7.5 %.
- Method : Ion-exchange high-performance liquid chromatography (HPLC).
Reference : American Diabetes Associations. Standards of Medical Care IN Diabetes 2023
- The presence of SOME hemoglobinopathies/hemoglobin variants may be missed BY SOME platforms used FOR measuring HbA1c values. IN these cases, the HbA1c result may vary OR give deviant results, depending ON the platform used. Alternative measures of glycemic control LIKE fructosamine OR glycated albumin may be considered IN such cases.
 - In presence of hemoglobinopathy/hemoglobin variants IN homozygous/compound heterozygous conditions alternative forms of testing such AS fructosamine OR glycated albumin may be considered.

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No:D0257183

Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Sodium Fluoride

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:28AM
Report Generated On : 12/Sep/2026 11:01AM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Fasting Blood Sugar

Glucose, Fasting	78.5	mg/dl	70 - 100
------------------	------	-------	----------

Method: Hexokinase
Machine: BECKMAN COULTER DXC 700 AU

American Diabetes Association Reference Range :

Normal : < 100 mg/dl
Impaired fasting glucose(Prediabetes) : 100 - 125 mg/dl
Diabetes : >= 126 mg/dl

Conditions that can result in an elevated blood glucose level include:

- Diabetes mellitus ,Hemochromatosis ,Cushing syndrome ,Acromegaly and gigantism.
- Increased circulating epinephrine such as in pheochromocytoma and adrenalin injections
- Acute pancreatitis
- Chronic pancreatitis

Conditions that cause low blood glucose level include :

- Pancreatic disorders : Islet cell tumor , pancreatitis
- Hepatic disease (diffuse severe disease)
- Endocrine disorders : hypopituitarism, Addison's disease ,hypothyroidism
- Alcoholism
- Malnutrition

Preeti

Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs



SIN No:D0257183



Page 2 of 16

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:35AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 12:01PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Serum	Report Status	: Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
Lipid Profile			
Total Cholesterol Method: CHO-POD (Trac. Abel-Kendall) Machine: BECKMAN COULTER AU 5800	216.2	mg/dL	Desirable : <200 Borderline: 200-239 High : >=240
Serum Triglycerides Method: GPO-POD Machine: BECKMAN COULTER AU 5800	216.7	mg/dL	Desirable : <150 Borderline high : 150-199 High : 200-499 Very high : >= 500
Serum HDL Cholesterol Method: ENZYMATIC IMMUNOINHIBITION Machine: BECKMAN COULTER AU 5800	58.8	mg/dL	40 - 60
LDL Cholesterol Calculated Method: Calculated	114.06	mg/dL	Optimal : <100 near /above Optimal:100 - 129 Borderline High: 130- 159 High : 160 - 189 Very High :>=190
VLDL Cholesterol Calculated Method: Calculated	43.34	mg/dl	<30
Total CHOL / HDL Cholesterol Ratio Method: Calculated	3.68	Ratio	3.30 - 4.40
LDL / HDL Cholesterol Ratio Method: Calculated	1.94	Ratio	Desirable/Low Risk: 0.5-3.0 Line/Moderate Risk: 3.0-6.0 Elevated/High Risk: >6.0
HDL / LDL Cholesterol Ratio Method: Calculated	0.52	Ratio	Optimal->0.4 Moderate-0.4 to 0.3 High-<0.3
Non-HDL Cholesterol Method: Calculated	157.4	mg/dL	0.0 - 160.0

Dyslipidemia is a disorder of fat or lipoprotein metabolism in the body and includes lipoprotein overproduction or deficiency. Dyslipidemias means increase in the level of one or more of the following: Total Cholesterol, low density lipoprotein (LDL) and/or triglyceride concentrations. Dyslipidemia also includes a decrease in the "good" cholesterol or high-density lipoprotein (HDL) concentration in the blood. Cholesterol is a steroid carried in the bloodstream as lipoprotein, necessary for cell membrane functioning and as a precursor to bile acids, progesterone ,vitamin D ,estrogens ,glucocorticoids and mineralocorticoids. HDL is termed "good cholesterol" because its levels are inversely related to the risk of Coronary heart disease.

Preeti
Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs


 SIN No:D0257183

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:35AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 12:01PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Serum	Report Status	: Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

LDL cholesterol is termed the "bad cholesterol" and their increased levels are associated with increased risk of atherosclerosis and coronary heart disease. Lipid level assessments must be made following 9 to 12 hours of fasting, otherwise assay results might lead to erroneous interpretation. Healthians labs report biological reference intervals (normal ranges) in accordance with the recommendations of The National Cholesterol Education Program (NCEP) & Adult Treatment Panel IV (ATP IV) guidelines providing the most desirable targets of various circulating lipid fractions in the blood. NCEP recommends that all adults above 20 years of age must be screened for abnormal lipid levels.

Updated risk stratification approach recommended by the Lipid Association of India (2023)			
Risk Category			
Extreme Risk	Category A: ASCVD with ≥ 1 feature of high-risk group •CACS ≥300 •Homozygous FH		
	Category B: ASCVD with ≥ 1 feature of very high-risk group or • Recurrent ACS or • Polyvascular disease or •Homozygous FH		
	Category C: Recurrent ASCVD event despite LDL-C around 30 mg/dL. These patients require special consideration.		
Very High Risk (VHR)	•Diabetes with target organ damage • Diabetes with ≥2 major ASCVD risk factors •CACS 100–299 or >75 percentile if CACS 1–99 •≥2 high risk features •Established ASCVD (obstructive or non-obstructive) •Heterozygous FH or LDL-C ≥190 mg/dL		
High Risk (HR)	•>3 major ASCVD risk factors, or •LDL-C 160–189 mg/dL or •Non-HDL-C 190–219 mg/dL or •Diabetes with 0–1 major ASCVD risk factor •2 major ASCVD factors + ≥1 risk modifier or •Any 1 high-risk feature.		
Moderate Risk (MR)	•2 major ASCVD risk factors, or •LDL-C 130–159 mg/dL or •Non-HDL-C 160–189 mg/dL or •Low-risk group + ≥1 risk modifier or lifetime ASCVD risk >30%		
Low Risk (LR)	•0–1 major ASCVD risk factor and •LDL-C 100–129 mg/dL and •Non-HDL-C 130–159 mg/dL, and •Lifetime CVD risk <30%		
Major Risk ASCVD (Atherosclerotic Cardiovascular Disease) Risk Factors			
•Age ≥45 years in males and ≥55 years in females •Current cigarette smoking or tobacco use •High blood pressure •Low HDL-C			
Treatment targets for lipid lowering therapy for various atherosclerotic cardiovascular disease risk groups			
Risk Groups	Treatment Targets		
	LDL-C, mg/dl (Primary Target)	Non-HDL-C,mg/dl(Coprimary Target)	Apo-B, mg/dl (Secondary Target)
Low risk group	<100	<130	<90
Moderate risk group	<100 (optional <70)	<130 (optional <100)	<90
High risk group	<70	<100	<80
Very high risk	<50	<80	<65
Extreme -risk-group- Category (A)	<50 (optional ≤30)	<80 (optional ≤60)	<65
Extreme -risk-group- Category (B)	≤30	≤60	<50
Extreme -risk-group- Category (C)	10-15	40-45	-----

Reference: Journal of Clinical Lipidology, Vol-18 No-3. May/June 2024

Preeti
Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs


 SIN No:D0257183

Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Serum

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:35AM
Report Generated On : 12/Sep/2026 12:01PM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Liver Function Test (LFT)

Serum Bilirubin, (Total) Method: DPD Machine: BECKMAN COULTER AU 5800	0.77	mg/dl	0.3 - 1.2
Serum Bilirubin, (Direct) Method: DPD Machine: BECKMAN COULTER AU 5800	0.16	mg/dl	0 - 0.2
Serum Bilirubin, (Indirect) Method: Calculated	0.61	mg/dl	0.0 - 0.8
Aspartate Aminotransferase (AST/SGOT) Method: UV without P5P Machine: BECKMAN COULTER AU 5800	21.20	U/L	3- 50
Alanine Aminotransferase (ALT/SGPT) Method: UV without P5P Machine: BECKMAN COULTER AU 5800	25.6	U/L	3 - 50
Alkaline Phosphatase (ALP) Method: AMP Buffer Machine: BECKMAN COULTER AU 5800	69.00	U/L	43-115
Gamma Glutamyl Transferase (GGT) Method: IFCC Machine: BECKMAN COULTER AU 5800	19.0	U/L	5 -55
Serum Total Protein Method: BIURET Machine: BECKMAN COULTER AU 5800	7.58	g/dl	6.6 - 8.3
Serum Albumin Method: BROMOCRESOL GREEN Machine: BECKMAN COULTER AU 5800	4.40	g/dl	3.5 - 5.2
Serum Globulin Method: Calculated	3.18	gm/dl	3.0 - 4.2
Albumin/Globulin Ratio Method: Calculated	1.38	Ratio	1.2 - 2.5
SGOT/SGPT Ratio Method: Calculated	0.83	Ratio	0.7 - 1.4

Bilirubin is a yellowish pigment found in bile and is a breakdown product of normal heme catabolism. Elevated levels are a result of increased bilirubin production (e.g hemolysis and ineffective erythropoiesis), decreased bilirubin excretion (e.g.; obstruction and hepatitis) and abnormal bilirubin metabolism (e.g; hereditary and neonatal jaundice) .

Preeti
Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs

SIN No:D0257183



Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:35AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 12:01PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Serum	Report Status	: Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Conjugated (direct) bilirubin is elevated in conditions like- Hereditary disorders(Dubin Johnson syndrome, Rotor syndrome),Hepatocellular damage(e.g –viral ,toxic ,alcohol ,drugs) ,biliary duct obstruction (extrahepatic or intrahepatic), Infiltrations ,space occupying lesions(e.g metastasis, abscess , granuloma , amyloidosis. Increased unconjugated (indirect) bilirubin may be a result of hemolytic or pernicious anemia, transfusion reaction & a common metabolic condition termed Gilbert syndrome.

AST levels increase in viral hepatitis, blockage of the bile duct ,cirrhosis of the liver, liver cancer, kidney failure, hemolytic anemia, pancreatitis, hemochromatosis. AST levels may also increase after a heart attack or strenuous activity.

ALT is a liver specific enzyme commonly measured as a part of a diagnostic evaluation of hepatocellular injury, to determine liver health.

Elevated ALP levels are seen in Biliary Obstruction, Osteoblastic Bone Tumors, Osteomalacia, Hepatitis, Hyperparathyroidism, Leukemia, Lymphoma, Paget's disease, Rickets, Sarcoidosis etc.

Elevated serum GGT activity can be found in diseases of the liver, Biliary system and pancreas. Obstructive liver disease, high alcohol consumption and use of enzyme-inducing drugs lead to raised GGT levels .

Serum total protein measures the total amount of protein in serum. It is largely comprised of albumin and globulins. Increased levels may be due to: Chronic inflammation or infection, including HIV and hepatitis B or C, multiple myeloma, Waldenstrom's disease. Decreased levels may be due to: Agammaglobulinemia, Bleeding (hemorrhage), Burns, Glomerulonephritis, Liver disease, Malabsorption, Malnutrition, Nephrotic Syndrome.

Albumin is the most abundant protein in the serum and is produced in the liver. Low serum albumin levels (hypoalbuminemia) can be caused by: Liver diseases like liver cirrhosis, nephrotic syndrome, protein-losing enteropathy, burns, hemodilution, increased vascular permeability or decreased lymphatic clearance, malnutrition and wasting .

Globulins are increased in most liver diseases , in chronic inflammatory diseases and neoplastic diseases.

Please note that there is a change in the Biological Reference Interval for Alkaline Phosphatase (ALP) parameter. (Dated: 11.02.2026).

(KFT) Kidney Function Test

Serum Creatinine	0.91	mg/dl	0.6 - 1.6
Method: Modified Jaffe, Kinetic			
Machine: BECKMAN COULTER AU 5800			
GFR, ESTIMATED	111.32	mL/min/1.73m2	
Method: Calculated (CKD-EPI)			
Serum Uric Acid	5.9	mg/dl	3.5-7.2
Method: Uricase PAP			
Machine: BECKMAN COULTER AU 5800			
Serum Calcium	10.0	mg/dl	8.8 - 10.6
Method: ARSENAZO III			
Machine: BECKMAN COULTER AU 5800			
Serum Sodium	139	mmol/L	136 - 146
Method: ISE (Indirect)			
Machine: BECKMAN COULTER AU 5800			
Serum Chloride	102	mmol/L	101 - 109
Method: ISE (Indirect)			

Preeti

Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs



SIN No:D0257183

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:35AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 12:01PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Serum	Report Status	: Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name	Value	Unit	Bio. Ref Interval
Machine: BECKMAN COULTER AU 5800			
Blood Urea	20.6	mg/dl	17 - 43
Method: GLDH,Kinetic assay			
Machine: BECKMAN COULTER AU 5800			
Blood Urea Nitrogen (BUN)	9.6	mg/dl	8 - 20
Method: Calculated			
Bun/Creatinine Ratio	10.58	Ratio	
Method: Calculated			
Urea/Creatinine Ratio	22.64	Ratio	
Method: Calculated			

The kidneys play a vital role in the excretion of waste products and toxins such as urea, creatinine and uric acid, regulation of extracellular fluid volume, serum osmolality and electrolyte concentrations, as well as the production of hormones like erythropoietin and 1,25 dihydroxy vitamin D and renin. Assessment of renal function is important in the management of patients with kidney disease or pathologies affecting renal function. Tests of renal function have utility in identifying the presence of renal disease, monitoring the response of kidneys to treatment, and determining the progression of renal disease.

Urea is synthesized in the liver as the final product of protein and amino acid metabolism. Urea synthesis is therefore dependent on daily protein intake and endogenous protein metabolism.

Creatinine is a metabolic product of creatine and phosphocreatine, which are both found almost exclusively in muscle.

Uric Acid is the major product of purine catabolism in humans. Uric acid levels are used to monitor the treatment of gout.

Measurement of calcium is used in the diagnosis and treatment of parathyroid disease, a variety of bone diseases, chronic renal disease, urolithiasis and tetany. Phosphorus levels are increased in acute or chronic renal failure with decreased GFR .

Sodium is an electrolyte, and it helps regulate the amount of water in and around the cells & the balance of chemicals in the body called acids and bases. Potassium is a primary intracellular ion, only 2 % is extracellular, high intracellular concentration is maintained by a Na- K ATPase pump, which continuously transports potassium into the cell against a concentration gradient. Chloride is a negatively charged ion that works with other electrolytes such as potassium, sodium, and bicarbonate, to help regulate the amount of fluid in the body and maintain the acid-base balance.

EGFR-National kidney Disease Education program recommends the use of MDRD equation to predict/ estimate GFR in adults (>=20 years) with Chronic kidney disease. Calculated by serum creatinine may be less accurate due to certain factor like race muscle mass, diet and certain drugs in such case EGFR should be calculated using serum cystatin C.

Age (Years)	Male ML/min/1.73m ²	Female ML/min/1.73m ²
17-24	93-131	—
25-34	78-146	—

Preeti
Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs



SIN No: D0257183



Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Serum

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:35AM
Report Generated On : 12/Sep/2026 12:01PM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF BIOCHEMISTRY

Test Name Value Unit Bio. Ref Interval

35-44	74-138	65-123
45-54	74-129	58-110
55-64	69-122	50-110
65-74	61-114	46-105
75-84	52-102	48-85

Preeti

Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs



SIN No:D0257183



Page 8 of 16

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:35AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 11:59AM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Serum	Report Status	: Final Report

DEPARTMENT OF IMMUNOLOGY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Thyroid Stimulating Hormone (TSH)

Thyroid Stimulating Hormone (TSH)-Ultrasensitive	2.554	µIU/ml	0.38-5.33
Method: CLIA			
Machine: Beckman Coulter UniCel DxI800			

Interpretation:

Pregnancy interval	Bio Ref Range for TSH in uIU/ml (As per American Thyroid Association)
First trimester	0.1 - 2.5
Second trimester	0.2 - 3.0
Third trimester	0.3 - 3.0

1. Thyroid hormones undergo circadian variation: Peak levels are seen at around midnight. Minimum levels seen between morning through noon. This variation may be as much as 50% thus, influence of sampling time needs to be considered for clinical interpretation.
2. Circulating forms of T3 and T4 are mostly reversibly bound with Thyroxine binding globulins (TBG), and to a lesser extent with albumin and Thyroid binding Pre-Albumin. Thus the conditions in which TBG and protein levels alter such as chronic liver disorders, pregnancy, excess of estrogens, androgens, anabolic steroids and glucocorticoids may cause misleading total T3, total T4 and TSH interpretations.
3. TSH levels may be normal in central hypothyroidism, recent rapid correction of hypothyroidism or hyperthyroidism, pregnancy, phenytoin therapy etc.
4. TSH values of <0.03 uIU/mL must be clinically correlated to evaluate the presence of a rare TSH variant in certain individuals which is undetectable by conventional methods.
5. Presence of Autoimmune disorders may lead to spurious results of thyroid hormones.
6. Various drugs influence the levels of thyroid hormones such as L-Dopa, Lithium, Glucocorticoids, Phenytoin etc.
7. Serum TSH is evaluated in Neonates to diagnose Congenital Hypothyroidism.
8. Within hours of birth, plasma TSH, T4, and T3 concentrations rise rapidly. By 2 to 3 days, TSH concentrations fall. T4 falls to adult concentrations by 1 to 2 months of age.

Vitamin B12

VITAMIN B12	207	pg/mL	222 – 1439
Method: CLIA			
Machine: Beckman Coulter UniCel DxI800			

Interpretation

Vitamin B12 is a coenzyme that is involved in two very important metabolic functions vital to normal cell growth and DNA synthesis: 1) the synthesis of methionine from homocysteine and 2) the conversion of methyl malonyl CoA to succinyl CoA. Deficiency of this vitamin can lead to megaloblastic anemia and ultimately to severe neurological problems. It can also lead to macrocytic anemia, glossitis, peripheral neuropathy, weakness, hyperreflexia, ataxia, loss of proprioception, poor coordination, and affective behavioral changes. A significant increase in RBC mean corpuscular volume (MCV) may be an important indicator of vitamin B12 deficiency. Patients taking vitamin B12 supplementation may have misleading results. A normal serum concentration of Vitamin B12 does not rule out tissue deficiency of vitamin B12. The most sensitive test for Vitamin B12 deficiency at the cellular level is the assay for methyl malonic acid (MMA). If clinical symptoms suggest deficiency, measurement of MMA and homocysteine should be considered, even if serum B12 concentrations are normal.

The Bio Ref Interval for Vitamin B12 has been changed on Beckman Coulter UniCel DxI800 with effect from 18.2.25.

Vitamin D, 25-Hydroxy

VITAMIN D (25 - OH VITAMIN D)	15.25	ng/ml	30-100
-------------------------------	-------	-------	--------

Preeti

Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs



SIN No:D0257183

Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Serum

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:35AM
Report Generated On : 12/Sep/2026 11:59AM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF IMMUNOLOGY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Method: CLIA

Machine: Beckman Coulter UniCel DxI800

VITAMIN D STATUS	VITAMIN D 25 HYDROXY (ng/mL), Adult	VITAMIN D 25 HYDROXY (ng/mL), Pediatric
DEFICIENCY	<20	<15
INSUFFICIENCY	20 - 30	15 - 20
SUFFICIENCY	30 - 100	20 - 100

Vitamin D is a steroid hormone known for its important role in regulating body levels of calcium and phosphorus and in the mineralization of bone

Uses:

- Diagnosis of Vitamin D deficiency
- Differential diagnosis of causes of rickets and Osteomalacia
- Monitoring Vitamin D replacement therapy
- Diagnosis of hypervitaminosis D

Increased in:

- Vitamin D intoxication
- Excessive exposure to sunlight

Decreased in:

- Malabsorption
- Steatorrhoea
- Dietary osteomalacia
- Thyrotoxicosis
- Coeliac disease
- Inflammatory bowel disease
- Rickets
- Pancreatic insufficiency

Preeti

Dr. Preeti
MBBS, MD Biochemistry
Consultant Biochemist
DMC - 97534, Healthians Labs



SIN No: D0257183



Page 10 of 16

Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Whole Blood EDTA

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:55AM
Report Generated On : 12/Sep/2026 01:25PM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF HAEMATOLOGY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Complete Blood Count

Haemoglobin (HB) Method: Spectrophotometry Machine: Yumizen H1500	14.8	g/dL	13.0-17.0
Total Leucocyte Count (TLC) Method: Impedance Machine: Yumizen H1500	5.7	10 ³ /uL	4.0-10.0
Hematocrit (PCV) Method: Calculated Machine: Yumizen H1500	44.9	%	40.0-50.0
Red Blood Cell Count (RBC) Method: Impedance Machine: Yumizen H1500	4.97	10 ⁶ /μl	4.50-5.50
Mean Corp Volume (MCV) Method: Derived from RBC Histogram Machine: Yumizen H1500	90.3	fL	83.0-101.0
Mean Corp Hb (MCH) Method: Calculated Machine: Yumizen H1500	29.8	pg	27.0-32.0
Mean Corp Hb Conc (MCHC) Method: Calculated Machine: Yumizen H1500	33.1	g/dL	31.5-34.5
RDW - CV Method: Derived from RBC Histogram Machine: Yumizen H1500	14.0	%	11.6-14.0
RDW - SD Method: Derived from RBC Histogram Machine: Yumizen H1500	39.20	fL	39.0-46.0
Mentzer Index Method: Calculated	18.17	Ratio	
RDWI Method: Calculated	254.37	Ratio	
Green and king index Method: Calculated	77	Ratio	

Differential Leucocyte Count

Neutrophils	47.7	%	40 - 80
-------------	------	---	---------

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No: D0257183

Patient Name : Mr Prakhar Srivastava
Age/Gender : 37Y 1M 29D /Male
Order Id : 20193635525
Referred By : Self
Customer Since : 12/Sep/2026
Sample Type : Whole Blood EDTA

Barcode : D0257183
Sample Collected On : 12/Sep/2026 07:41AM
Sample Received On : 12/Sep/2026 10:55AM
Report Generated On : 12/Sep/2026 01:25PM
Sample Temperature : Maintained ✓
Report Status : Final Report

DEPARTMENT OF HAEMATOLOGY

Test Name	Value	Unit	Bio. Ref Interval
Method: Flow-Cytometry DHSS Machine: Yumizen H1500			
Lymphocytes	42.5	%	20-40
Method: Flow-Cytometry DHSS Machine: Yumizen H1500			
Monocytes	6.4	%	02 - 10
Method: Flow-Cytometry DHSS Machine: Yumizen H1500			
Eosinophils	3.3	%	01 - 06
Method: Flow-Cytometry DHSS Machine: Yumizen H1500			
Basophils	0.1	%	00 - 02
Method: Impedance Machine: Yumizen H1500			
Absolute Leucocyte Count			
Absolute Neutrophil Count (ANC)	2.72	10 ³ /uL	2.0-7.0
Method: Calculated			
Absolute Lymphocyte Count (ALC)	2.42	10 ³ /uL	1.0-3.0
Method: Calculated			
Absolute Monocyte Count	0.36	10 ³ /uL	0.2-1.0
Method: Calculated			
Absolute Eosinophil Count (AEC)	0.19	10 ³ /uL	0.02-0.5
Method: Calculated			
Absolute Basophil Count	0.01	10 ³ /uL	0.02 - 0.10
Method: Calculated			
Platelet Count(PLT)	186	10 ³ /μl	150-410
Method: Impedance Machine: Yumizen H1500			
MPV	11.0	fL	7 - 9
Method: Derived from PLT Histogram Machine: Yumizen H1500			

The International Council for Standardization in Haematology (ICSH) recommends reporting of absolute counts of various WBC subsets for clinical decision making. This test has been performed on a fully automated 5 part differential cell counter which counts over 10,000 WBCs to derive differential counts. A complete blood count is a blood panel that gives information about the cells in a patient's blood, such as the cell count for each cell type and the concentrations of Hemoglobin and platelets. The cells that circulate in the bloodstream are generally divided into three types: white blood cells (leukocytes), red blood cells (erythrocytes), and platelets (thrombocytes). Abnormally high or low counts may be physiological or may indicate disease conditions, and hence need to be interpreted clinically.

The Mentzer index is used to differentiate iron deficiency anaemia beta thalassemia trait. If a CBC indicates microcytic anaemia, these are two of the most likely

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No: D0257183

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183	
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM	
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:55AM	
Referred By	: Self	Report Generated On	: 12/Sep/2026 01:25PM	
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓	
Sample Type	: Whole Blood EDTA	Report Status	: Final Report	

DEPARTMENT OF HAEMATOLOGY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

causes, making It necessary to distinguish between them.

If the quotient of the mean corpuscular volume divided by the red blood cell count is then 13, thalassemia is more likely. If the result is greater than 13, then iron deficiency anaemia is more likely.

Green and King Index used to differentiate IDA from thalassemia trait value >65 is likely to be Iron Deficiency Anemia and value <220 more likely to be Iron Deficiency Anemia and value <220 more likely to be Beta Thalassemia Trait.

Calculated parameters are either derived from Impedence measure, RBC pulse measurement, RBC/platelet histograms or formula derived.

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No: D0257183

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:55AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 01:05PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Whole Blood EDTA	Report Status	: Final Report

DEPARTMENT OF HAEMATOLOGY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

ERYTHROCYTE SEDIMENTATION RATE (ESR)

ESR	7	mm/1st hour	0-10
Method: Modified Westergren Method			

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No: D0257183

Healthians Labs (A Unit of Expedient Healthcare Marketing Pvt. Ltd.)

Plot 1 & 2, Udyog Vihar, Phase-4, Gurgaon, Haryana, Pincode-122016

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:55AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 01:05PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Urine	Report Status	: Final Report

DEPARTMENT OF CLINICAL PATHOLOGY

Test Name	Value	Unit	Bio. Ref Interval
-----------	-------	------	-------------------

Urine Routine & Microscopy Extended

PHYSICAL EXAMINATION

Colour	Pale Yellow		Pale Yellow
Method: Visual			
Volume	15.00	mL	
Method: Visual			
Appearance	Clear		Clear
Method: Visual			

CHEMICAL EXAMINATION

Specific Gravity	1.015		1.001 - 1.035
Method: Dipstick-Ion exchange			
pH	6.0		4.5 - 7.5
Method: Dipstick-Double indicator			
Glucose	Negative		Negative
Method: Dipstick-oxidase peroxidase			
Urine Protein	Negative		Negative
Method: Dipstick-Bromophenol blue			
Ketones	Negative		Negative
Method: Sodium nitroprusside			
Urobilinogen	Normal		Normal
Method: Dipstick-Ehrlichs Test			
Bilirubin	Negative		Negative
Method: Dipstick-Ehrlichs Test			
Nitrite	Negative		Negative
Method: Dipstick-Griess test			
Blood	Negative		Nil
Method: Dipstick-Peroxidase			
Leucocyte Esterase	Negative		Nil
Method: Dipstick- Esterase			

MICROSCOPIC EXAMINATION

Pus Cells	1-2	/HPF	0 - 5
Method: Microscopic Examination			
Epithelial cells	1-2	/HPF	0 - 5
Method: Microscopic Examination			

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No: D0257183

Patient Name	: Mr Prakhar Srivastava	Barcode	: D0257183
Age/Gender	: 37Y 1M 29D /Male	Sample Collected On	: 12/Sep/2026 07:41AM
Order Id	: 20193635525	Sample Received On	: 12/Sep/2026 10:55AM
Referred By	: Self	Report Generated On	: 12/Sep/2026 01:05PM
Customer Since	: 12/Sep/2026	Sample Temperature	: Maintained ✓
Sample Type	: Urine	Report Status	: Final Report

DEPARTMENT OF CLINICAL PATHOLOGY

Test Name	Value	Unit	Bio. Ref Interval
RBCs	Nil	/HPF	Nil
Method: Microscopic Examination			
Casts	Nil		Nil
Method: Microscopic Examination			
Crystals	Nil		Nil
Method: Microscopic Examination			
Bacteria	Absent		Absent
Method: Microscopic Examination			
Yeast Cell	Absent		Absent
Method: Microscopic Examination			
Others (Non Specific)	Nil		NIL
Method: Microscopic Examination			

The main indication for testing for glucose in urine is detection of unsuspected diabetes mellitus or follow-up of known diabetic patients. Renal glycosuria accounts for 5% of cases of glycosuria in general population.

Proteinuria can be seen in nephrotic syndrome, pyelonephritis, heavy metal poisoning, tuberculosis of kidney, interstitial nephritis, cystinosis, Fanconi syndrome, rejection of kidney transplant. Hemodynamic proteinuria is transient and can be seen in high fever, hypertension, heavy exercise, congestive cardiac failure, seizures, and exposure to cold. Post-renal proteinuria is caused by inflammatory or neoplastic conditions in renal pelvis, ureter, bladder, prostate, or urethra.

Ketonuria can be seen in uncontrolled Diabetes mellitus with ketoacidosis, Glycogen storage disorder, starvation, persistent vomiting in children, weight reduction program, fever in children, severe thyrotoxicosis, pregnancy and protein calorie malnutrition.

Presence of bilirubin in urine indicates conjugated hyperbilirubinemia (obstructive or hepatocellular jaundice). Bile salts along with bilirubin can be detected in urine in cases of obstructive jaundice. Normally about 0.5-4 mg of urobilinogen is excreted in urine in 24 hours. Therefore, a small amount of urobilinogen is normally detectable in urine. Increased urobilinogen in urine can be seen due to hemolysis, megaloblastic anemia and haemorrhage in tissues. Decreased urobilinogen can be seen in obstructive jaundice, reduction of intestinal bacterial flora, neonates and following antibiotic treatment. The presence of abnormal number of intact red blood cells in urine is called as hematuria. It implies presence of a bleeding lesion in the urinary tract. Hematuria can be seen in glomerular diseases like Glomerulonephritis, Berger's disease, lupus nephritis, Henoch-Schonlein purpura, non glomerular diseases like Calculus, tumor, infection, tuberculosis, pyelonephritis, hydronephrosis, polycystic kidney disease, trauma, after strenuous physical exercise, diseases of prostate (benign hyperplasia of prostate, carcinoma of prostate).

Nitrites are not present in normal urine. Ingested nitrites are converted to nitrate and excreted

in urine. If gram-negative bacteria (e.g. E.coli, Salmonella, Proteus, Klebsiella, etc.) are present in urine, they will reduce the nitrates to nitrites through the action of bacterial enzyme nitrate reductase. As E. coli is the commonest organism causing urinary tract infection, this test is helpful as a screening test for urinary tract infection.

Some organisms like Staphylococci or Pseudomonas do not reduce nitrate to nitrite and therefore in such infections nitrite test is negative.

Leucocyte esterase test detects esterase enzyme released in urine from granules of leucocytes. Thus the test is positive in pyuria.

*** End Of Report ***

Pranjali

Dr. Pranjali Sejwal
MBBS, MD Pathology
Consultant Pathologist
Reg. No.: H N-20567, Healthians Labs



SIN No: D0257183

Healthians Labs (A Unit of Expedient Healthcare Marketing Pvt. Ltd.)

Plot 1 & 2, Udyog Vihar, Phase-4, Gurgaon, Haryana, Pincode-122016

Terms & Conditions

1. Test results released pertain to the sample/specimen submitted by the customer.
2. Test results are dependent on the quality of the samples received by the Lab.
3. The tests are carried out in the lab with the presumption that the sample/specimen belongs to the patient named or identified in the bill/test request form/booking ID.
4. The laboratory reports are not diagnostic. The reported results are for information only, and are subject to clinical correlation and interpretation by the consulting physician.
5. Test results may show interlaboratory variations depending upon different analyzers, methodologies, corresponding kits, biological reference intervals, and physiological status of the customer.
6. No internal data shall be shared with the customer.
7. Liability of Healthians Labs for deficiency of services or other errors and omissions shall be limited to the fee paid by the customer for the relevant laboratory services only.
8. This report is not subject to use for any medico-legal purposes.
9. The company has the right to cancel those tests which are not possible as per the received sample or clinical reference or information provided.
10. The reported values of the specific analytes are subject to the maximum reportable range as per the analyzers & kits.
11. Some tests may be performed in partner labs (outsourced). Expedient Healthcare Marketing Private Limited (Healthians Labs) makes no express or implied warranties regarding the technical precision, absolute accuracy, or Turnaround Times (TAT) of tests performed in partnered labs. Expedient Healthcare Marketing Private Limited (Healthians Labs) explicitly disclaims all liability for financial losses, operational downtime, schedule disruptions, or damages resulting from test performed at partner labs (outsourced). The name of the partner labs (outsourced) shall be mentioned explicitly on the respective reports.